ESSENTIALS OF ARTIFICIAL INTELLIGENCE

INDIVIDUAL CONTRIBUTION SHEET

Student Name	KAUC Prabhashwara	**Registration No.**	D/BIS/24/0004
Degree / Intake	BSc (Hons) in Information System / Intake 41	**Group No.**	28
Project Title	Serendib Go – AI-Based Intelligent Online Bus Booking Management System		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
AI System Architecture & Coordination	Led the overall AI integration strategy, designed the decoupled 3-tier microservice architecture (Next.js - Laravel - Python FastAPI), and coordinated project deliverables across all 10 weeks.	System Architecture, Workflow Design, Git/GitHub	Stage 1, 2, 3 System Architecture Diagrams, Project Proposals & Final Reports
XGBoost Demand Prediction Model	Developed, trained, and evaluated the XGBoost Regressor model to predict passenger ticket demand per route segment based on departure time, day of week, and holiday flags.	Python 3.10+, XGBoost, Scikit-learn, Pandas	predict_demand.py script, train_demand_model.py, demand_model.pkl (R^2 = 0.912)
Local Ollama LLM Integration	Configured local execution of Meta's Qwen 2.5 (qwen2.5:3b	Ollama Runtime, Qwen 2.5 (qwen2.5:3b via	ollama.png terminal execution log, ollama

	via Ollama) (3B) model via Ollama on Port 11434, guaranteeing 100% offline functionality, zero cloud API costs, and full passenger data privacy.	Ollama) (3B), REST API, Context Injection	live.png CLI session proof
FastAPI Microservice Layer	Built the Python FastAPI backend service that receives prediction requests from Laravel and returns real-time demand levels and predicted passenger counts.	Python, FastAPI, JSON Payloads, REST Endpoints	Python CLI script, JSON API logs (`predicted_demand: 1.4`)

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature:  **Date:** 17/10/2026

Student Name	TMND Hemarathne	**Registration No.**	D/BIT/24/0070
Degree / Intake	BSc (Hons) in Information Technology / Intake 41	**Group No.**	28
Project Title	Serendib Go – AI-Based Intelligent Online Bus Booking Management System		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Next.js 16 Web Application UI	Designed and developed the reactive web application user interface using Next.js 16 and React 19, ensuring responsive accessibility across mobile and desktop devices.	Next.js 16, React 19, Tailwind CSS, JavaScript	Frontend source code repository, responsive dashboard UI components
Interactive 40-Seat Bus Layout UI	Built the custom SmartSeatMap.jsx visual seating chart (A1-J4) for Mahinda Express luxury buses, showing real-time seat states (Booked, Available, Released).	React State Management, SVG/CSS Grid Layouts	SmartSeatMap.jsx component source code, seat map browser screenshots
Segment Journey Selector & Demand Badges	Created the origin/destination journey segment selector dropdown (e.g., Colombo to Chilaw) and integrated live XGBoost 'Moderate/High Demand' badges on the booking screen.	Next.js App Router, DemandIndicator.jsx	1.png & 22.png UI screenshots showing live demand indicators

AI Assistant Chatbot Modal UI	Developed the floating ChatWidget.jsx modal component that interacts directly with the local Ollama Qwen 2.5 backend for real-time customer support.	React Modal Hooks, REST API Client	ChatWidget.jsx frontend component, interactive chat UI screenshots

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Student Signature:  **Date:** 17/10/2026

Student Name	**DAKHD Arachchi**	**Registration No.**	D/BSE/25/0009
Degree / Intake	BSc in Software Engineering (BSE) / Intake 42	**Group No.**	28
Project Title	Serendib Go – AI-Based Intelligent Online Bus Booking Management System		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Laravel 12 REST API Gateway	Engineered the main application backend using Laravel 12, handling business logic, user authentication, routing, and HTTP communication between frontend and AI services.	Laravel 12, PHP 8.2, Eloquent ORM, RESTful API	Laravel controllers (`PredictionController.php`, `AIChatController.php`)
MySQL 8.0 Relational Database Design	Structured the database schema across `bookings`, `segments`, `seat_allocations`, `segment_availability`, `ml_predictions_log`, and `chatbot_conversations` tables.	MySQL 8.0, Relational Database Modeling, Indexing	MySQL ER Diagram, database migration files, SQL schema dumps
SQL Interval Tree Overlap Query Engine	Developed custom SQL Interval Tree range overlap logic (`start_stop_index` to `end_stop_index`) to verify non-overlapping	SQL Query Optimization, Interval Tree Range Logic	SQL Interval Tree query scripts in Laravel repository, database performance logs

	seat availability and prevent double booking during segment reselling.		
Context Injection & RAG Service Layer	Implemented the backend RAG service layer (`ChatbotService.php`) that fetches digitized NTC fare tables from MySQL and injects context into Ollama prompts.	PHP, MySQL Context Extraction, Ollama HTTP Client	ChatbotService.php source code, context-injected prompt logs

Declaration

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Student Signature:  **Date:** 17/10/2026

Student Name	**PADCM Perera**	**Registration No.**	D/DBA/25/0018
Degree / Intake	BSc in Data Science & Business Analytics (DBA) / Intake 42	**Group No.**	28
Project Title	Serendib Go – AI-Based Intelligent Online Bus Booking Management System		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Official NTC Fare Schedule Digitization	Digitized official July 2026 National Transport Commission (NTC) luxury air-conditioned fare schedules into structured MySQL database matrices (e.g., Route 004, Route 001).	NTC Official Fare Schedules PDF, Data Preprocessing, MySQL Ingestion	AC Fares (Effect from 2026-07-06).pdf data mapping, database fare tables
K-Means Passenger Behavioral Clustering	Built and tuned the K-Means unsupervised clustering model (k=4) to classify traveler profiles (Daily Commuters, Weekend Travelers, Holiday Surge, Last-Minute Bookers) for inventory protection.	Python, K-Means (`sklearn.cluster`), StandardScaler	train_clustering_model.py, clustering_model.pkl, elbow plot charts
10,000 Booking Synthetic Simulation	Generated a synthetic benchmark dataset of 10,000 booking requests across 90 days to test	Python Data Generation, Monte Carlo Simulation	10,000 row simulation CSV, benchmark data processing scripts

	traditional whole-route booking against Serendib Go's segment reselling.		
Experimental Evaluation & Statistical Validation	Executed paired t-tests ($p < 0.05$) and evaluated KPIs proving seat utilization increased from 55.4% to 82.7%, passengers served rose +45.2%, and rejections dropped <5%.	SciPy, Matplotlib, Statistical Analysis, Paired t-tests	Simulation benchmark result tables, evaluation charts in Final Report

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Student Signature:  **Date:** 17/10/2026